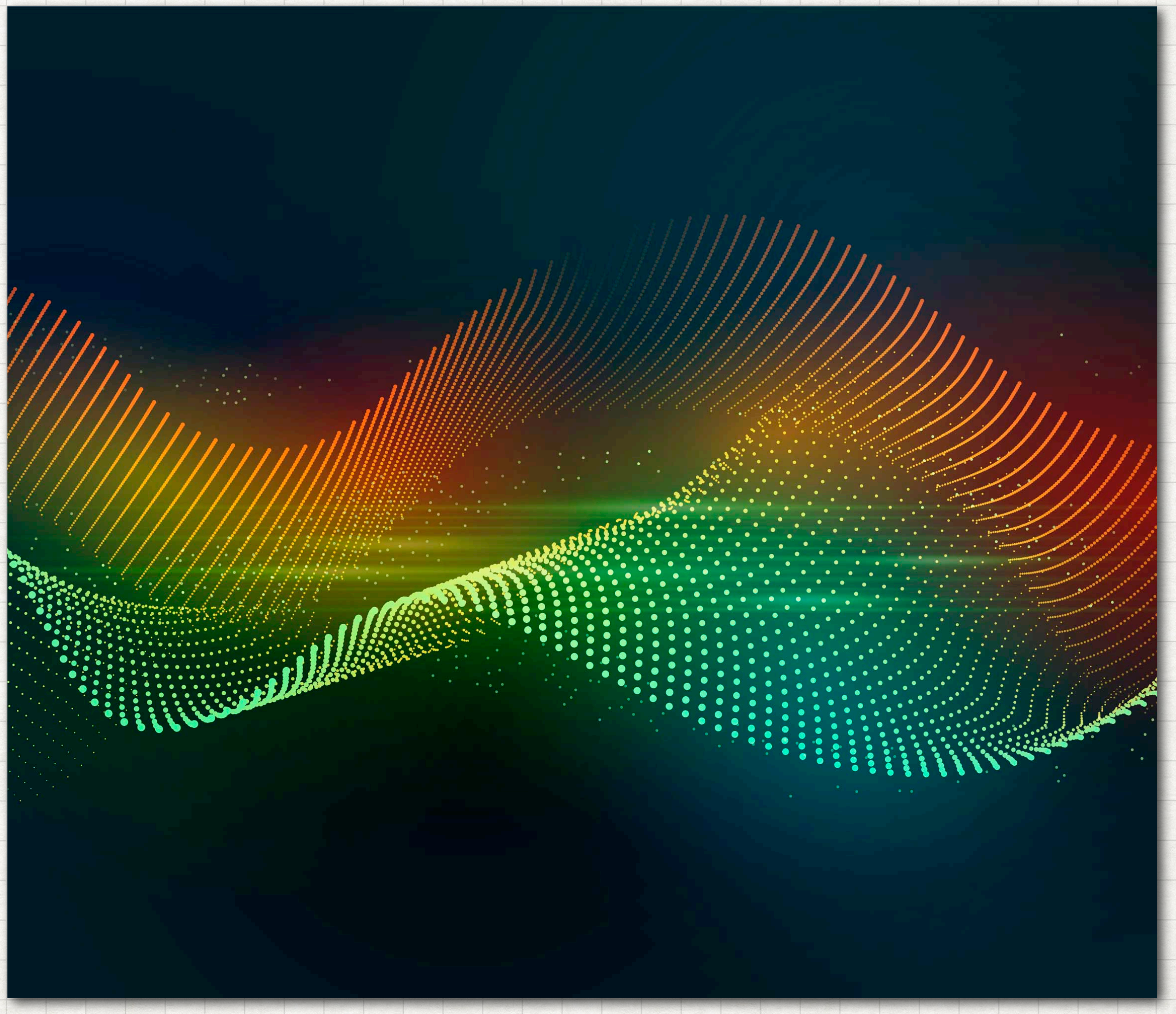
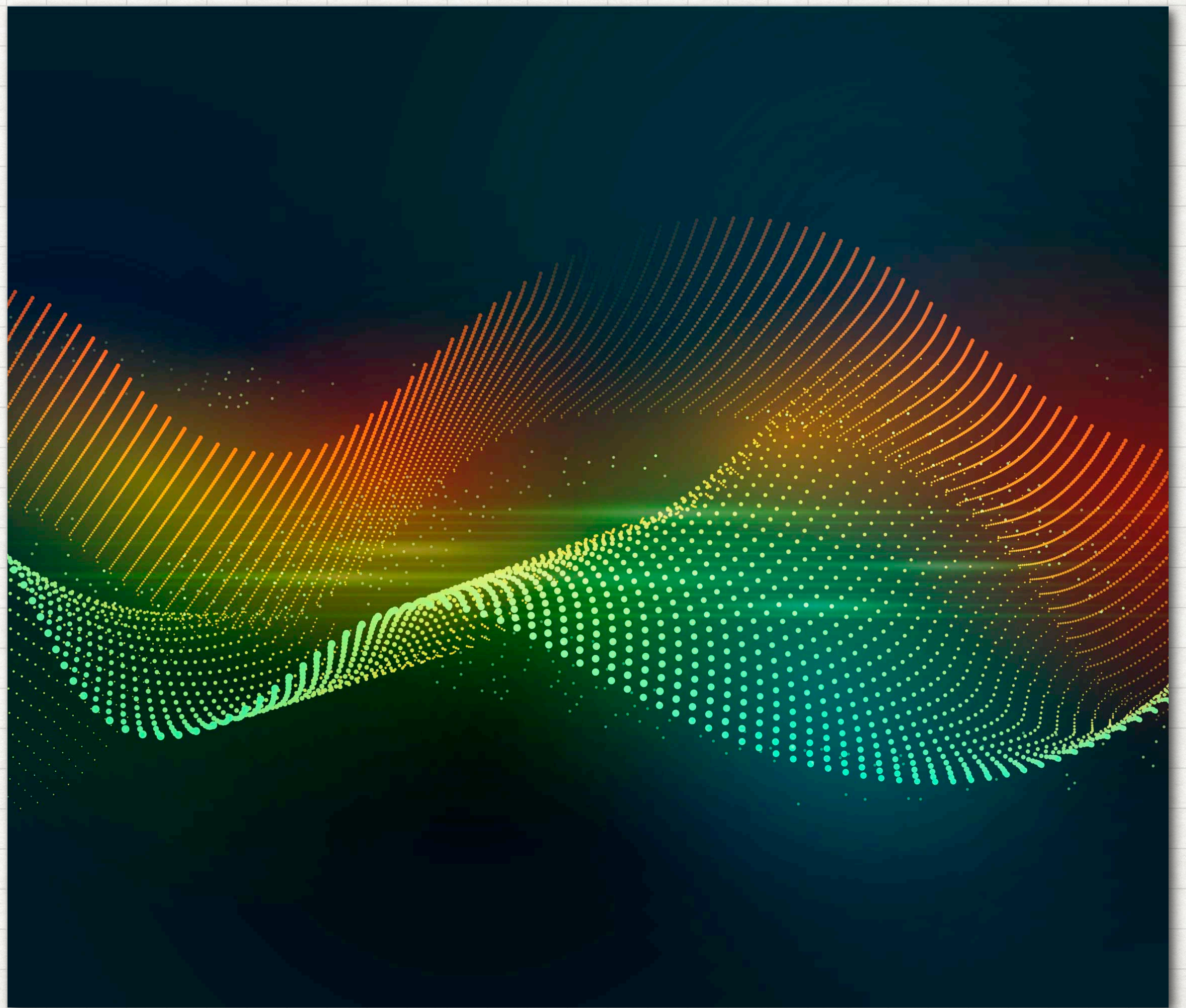
LECTURE 05

APPLICATIONS: FILTERING & OTHERS



LECTURE 05.01
SAMPLING



SAMPLING

- Continuous-time signal can be represented and continuously recoverable from knowledge of its values, or *samples* (discrete-time signal).

- Sampling theorem lies in its role as a bridge between continuous-time signals and discrete-time signals.

- Example, (1) moving pictures - consist of a sequence of individual frames viewed in sequence, accurately represented moving scene. (2) printed picture - spatial samples representing fine grid of point, the picture appears to be spatially continuous

- Processing discrete-time signals is more flexible and is often preferable to processing CT signals due to technology development, availability of inexpensive, lightweight, programmable, and easily reproducible discrete-time systems.

SAMPLING THEOREM

- Signals can have identical values at integer multiples of T
- $x_1(kT) = x_2(kT) = x_3(kT)$
- An infinite number of signals can generate a given set of samples.
- Sampling theorem: the samples are taken sufficiently close together in relation to the highest frequency present in the signal, then the samples uniquely specify the signal, and we can reconstruct it perfectly.

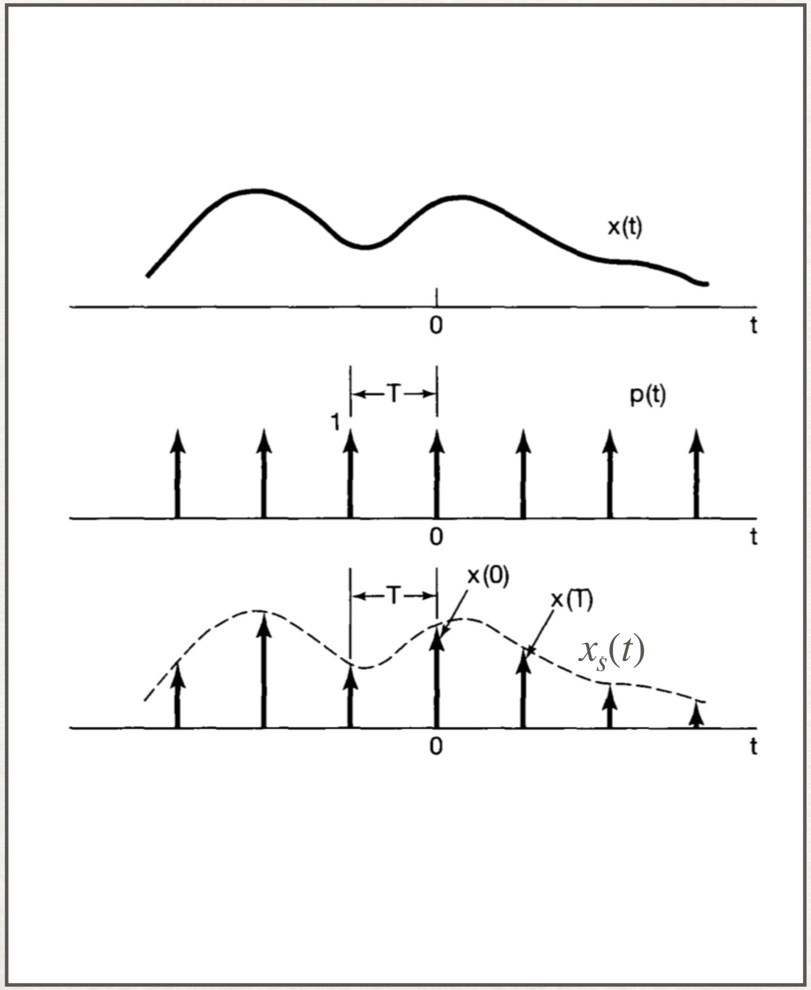
SAMPLING THEOREM

- Let $x(t)$ be a band-limited signal with $X(\omega) = 0$ for $|\omega| > \omega_B$ then $x(t)$ is uniquely determined by its samples $x(nT)$, $n = 0, \pm 1, \pm 2, \ldots$,if

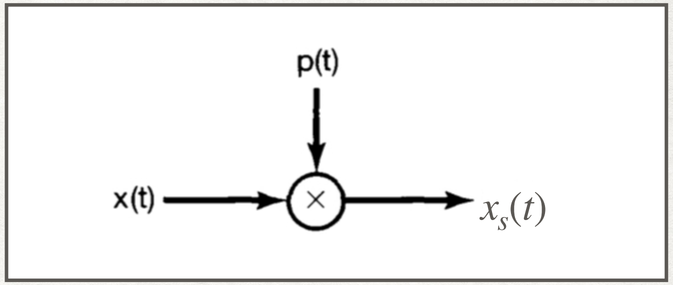
$$\omega_s \geq 2\omega_B, \quad \text{where } \omega_s = \frac{2\pi}{T}$$

- The sampling theorem allows us to completely reconstruct a band-limited signal x(t) from instantaneous samples taken at a rate $\omega_s = 2\pi/T$, provided that ω_s is at least as large as $2\omega_B$, which is twice the highest frequency present in the band-limited signal $x(t)$
- The minimum sampling rate $2\omega_B$ is known as the Nyquist rate
- The process of obtaining a set of samples from a continuous function of time $x(t)$ is referred to as sampling

IMPULSE TRAIN SAMPLING



- $x(t)$ is a low-pass signal (band limit) and no frequency > ω_B
- $p(t)$: periodic impulse train
- T_s : sampling period
- Output of the sampler $x_s(t)$



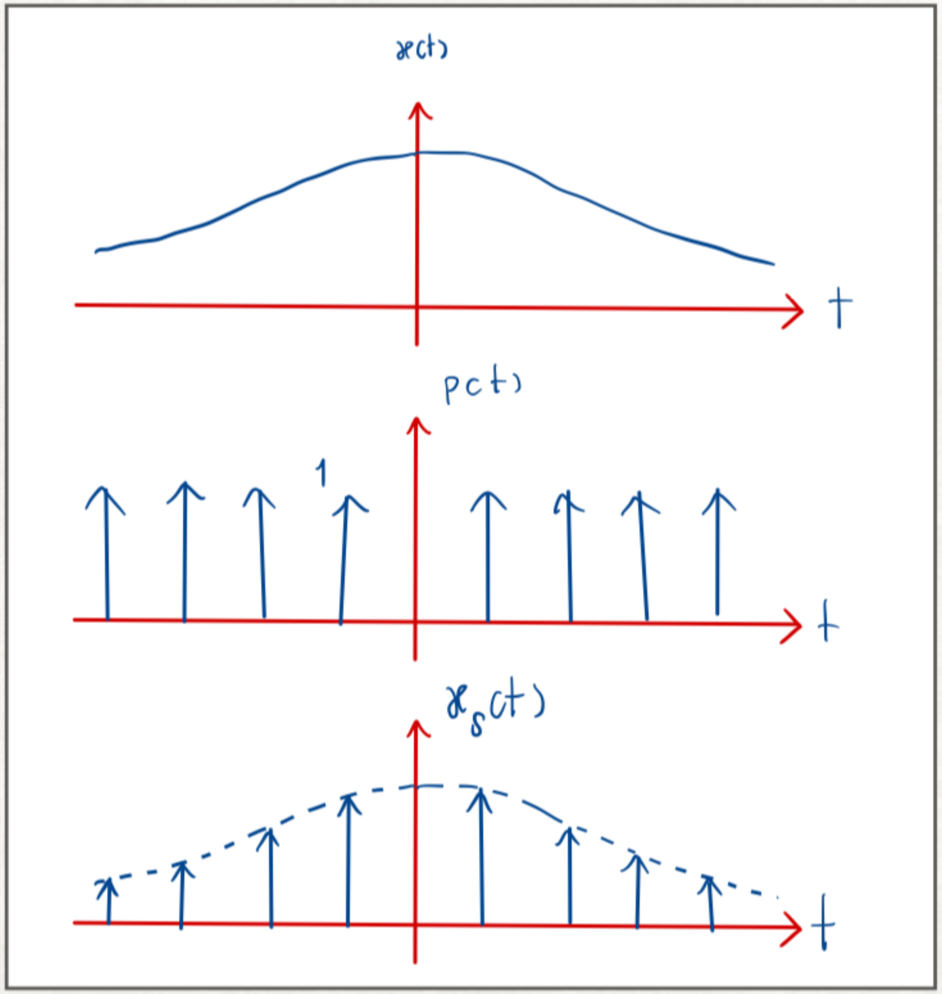
- Fundamental frequency (sampling f.) of $p(t)$ is $\omega_S = \dfrac{2\pi}{T_s}$

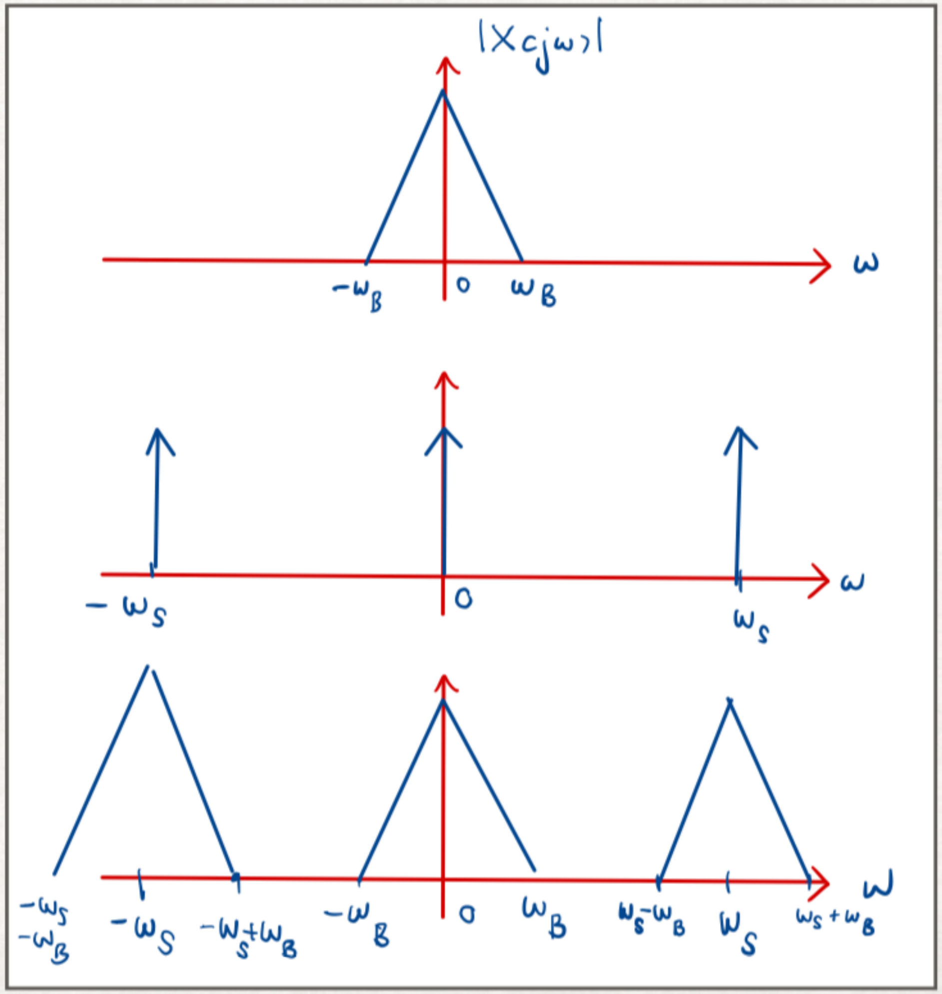
$$x_s(t) = x(t)p(t)$$

$$\text{where } p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$$

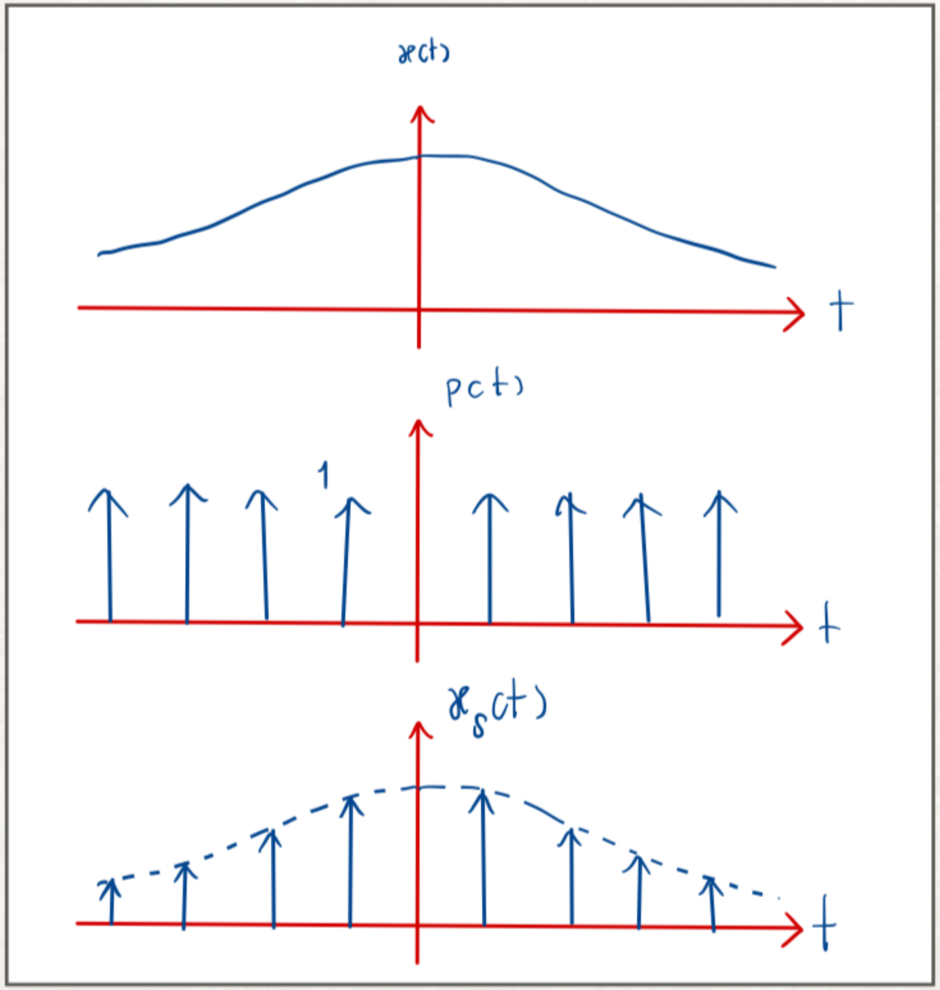
- The sampled signal is considered to be the product of a signal $x(t)$ and impulse train $p(t)$ —> impulse modulation model for sampling operations

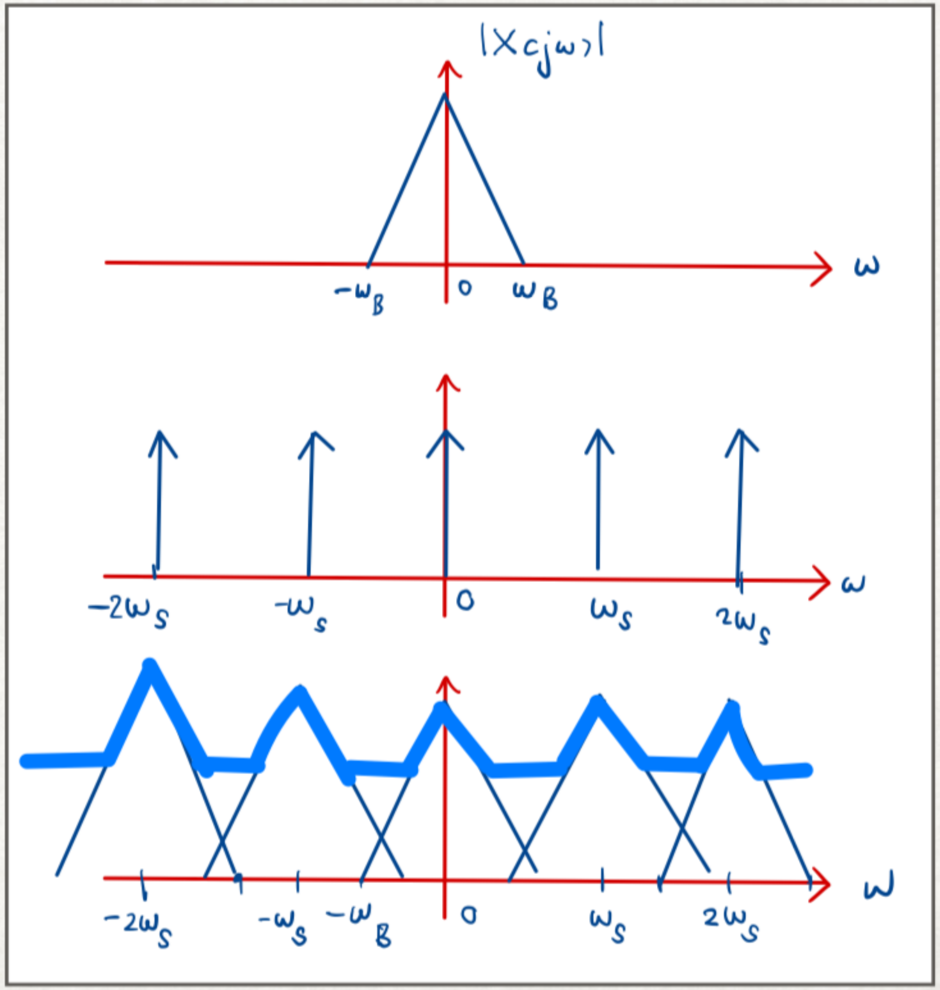
IMPULSE TRAIN SAMPLING





IMPULSE TRAIN SAMPLING





SAMPLING

EXERCISE

Spectrum of a signal, for example, speech is essentially zero for all frequencies above 5kHz. The Nyquist sampling rate for such a signal is….

ω_s = ______________________rad/s

T = _______________________ms (sample space)

EFFECT OF UNDER SAMPLING: ALIASING

- $\omega_s < 2\omega_B$, individual term overlap, referred to as "Aliasing"
- To calculate the perceived (reconstructed) frequency f_p of any signal frequency f, which is sampled at any sampling frequency f_s, can be calculated as
- $f_p = |f - kf_s|$ where k is the rounded value of $\dfrac{f}{f_s}$

EFFECT OF UNDER SAMPLING: ALIASING

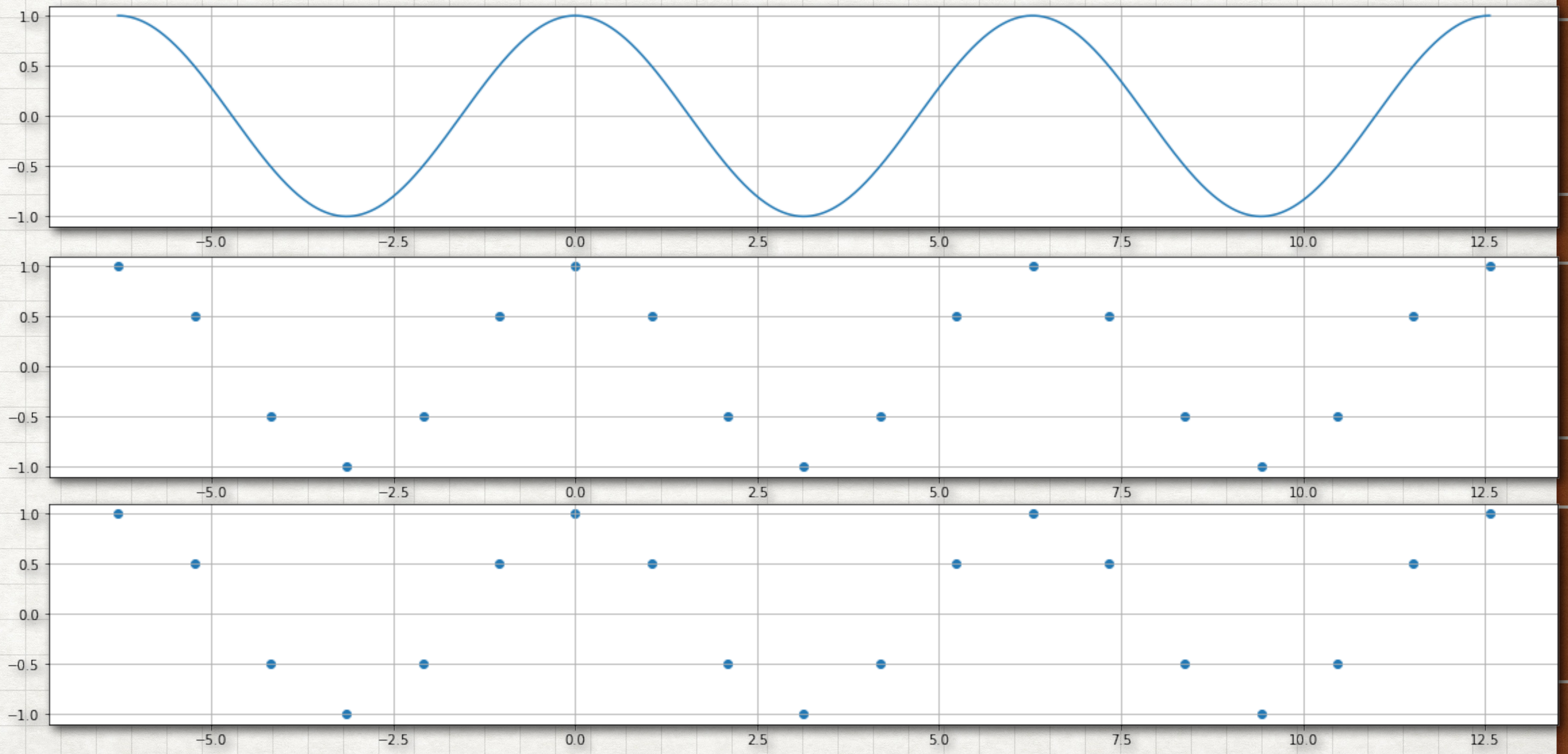
EXERCISE

Sketch the signal below and answer whether the sampling signal has "Aliasing"?

ω_0

$\omega_0 = \dfrac{\omega_S}{6}$

$\omega_0 = \dfrac{\omega_S}{3}$



EFFECT OF UNDER SAMPLING: ALIASING

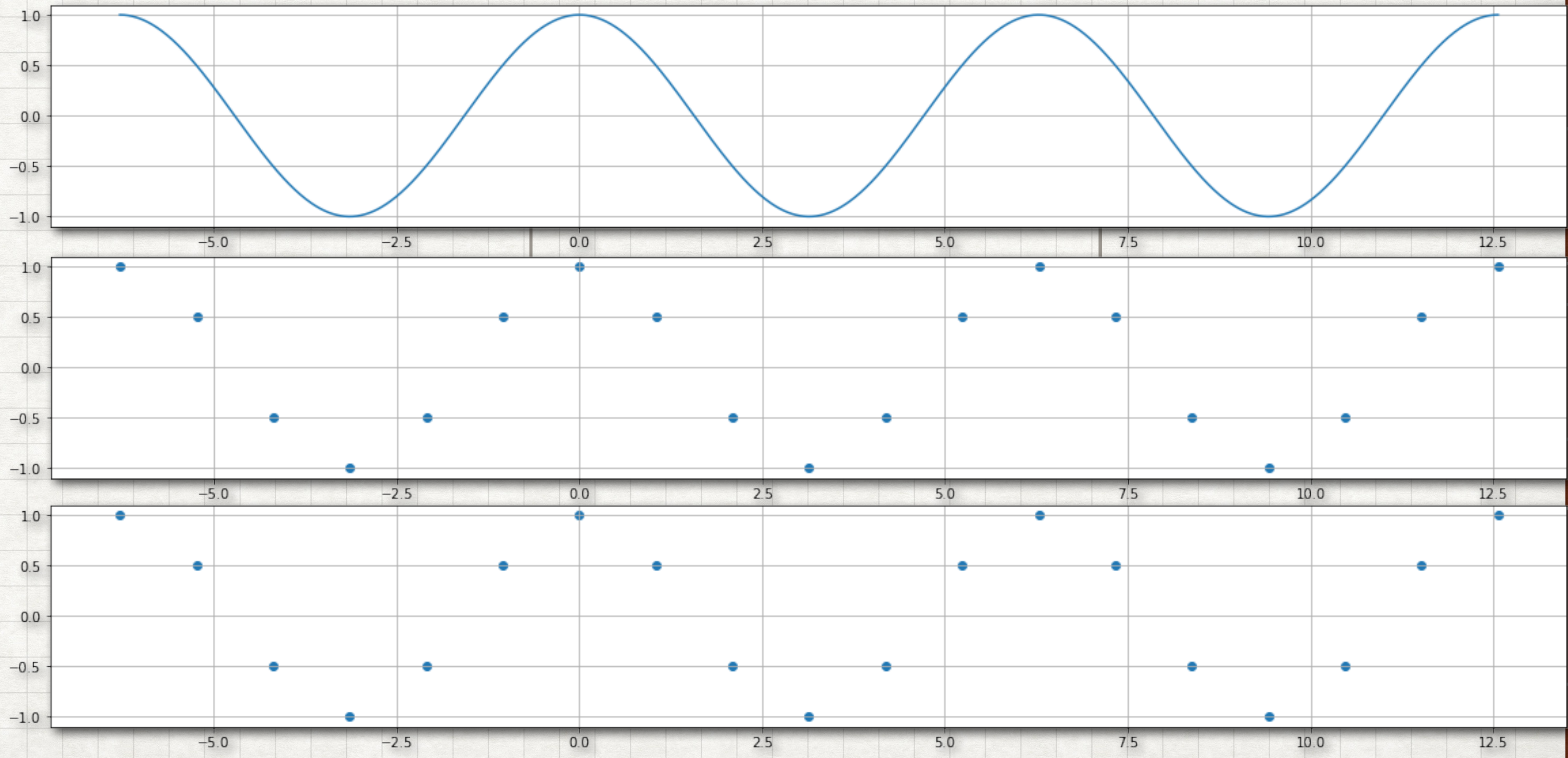
EXERCISE

Sketch the signal below and answer whether the sampling signal has “Aliasing”?

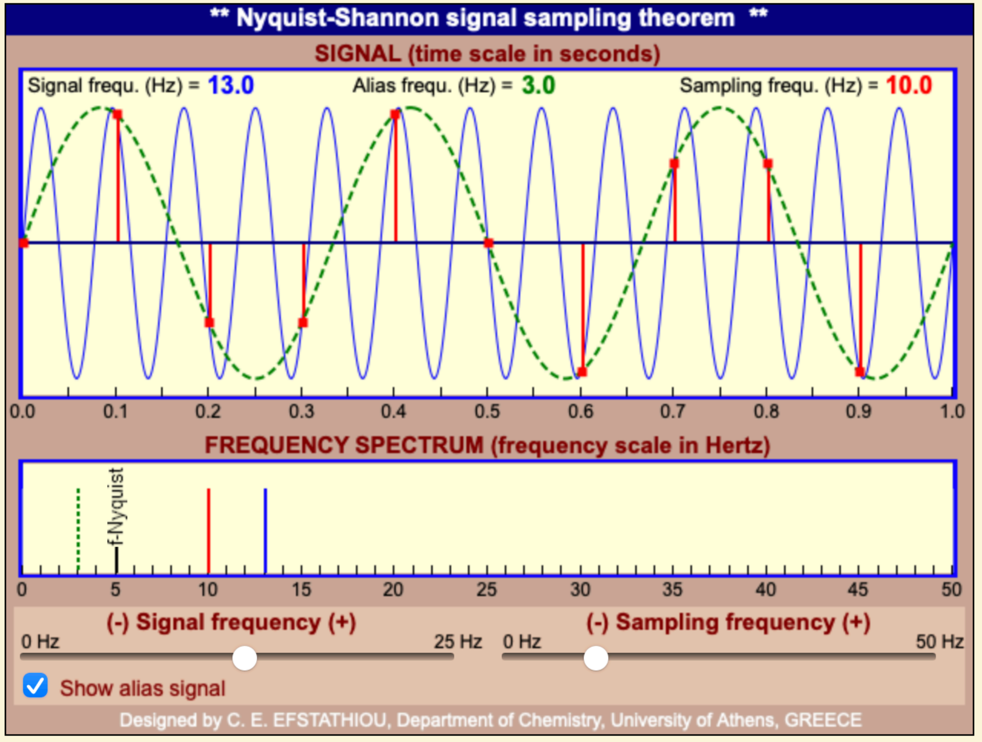
ω_0

$\omega_0 = \dfrac{4\omega_s}{6}$

$\omega_0 = \dfrac{5\omega_s}{6}$



DEMO



http://195.134.76.37/applets/AppletNyquist/Appl_Nyquist2.html

Page maintained by [Prof. C. E. Efstathiou]

REFERENCES

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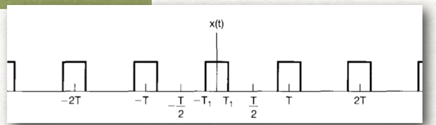
- Alan V. Oppenheim, Alan S. Willsky and S. Hamid Nawab., **Signals and Systems,** 2nd Edition, Prentice Hall (1996)
  - Chapter 7 (7.1,7.3)
- Samir S. Soliman and Mandyam D. Srinath., **Continuous and Discrete Signals and Systems**, 2nd Edition, Prentice Hall (1998)
  - Chapter 4 (4.4.3)

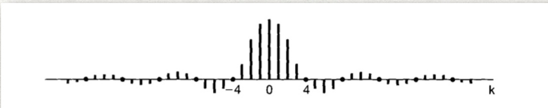
COMPARISON

FOURIER SERIES

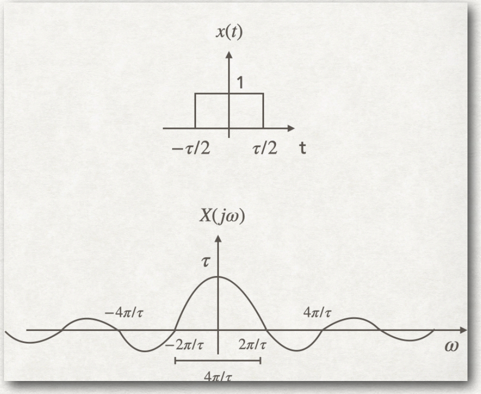


$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} = \sum_{k=-\infty}^{\infty} a_k e^{jk(2\pi/T)t}$$

$$a_k = \frac{1}{T}\int_{\langle T\rangle} x(t)e^{-jk\omega_0 t}\,dt = \frac{1}{T}\int_0^T x(t)e^{-jk(2\pi/T)t}dt$$



FOURIER TRANSFORM



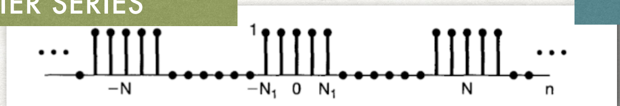
$$x(t) = \frac{1}{2\pi}\int_{\omega=-\infty}^{\infty} X(j\omega)e^{j\omega t}d\omega$$

$$X(j\omega) = \int_{t=-\infty}^{\infty} x(t)e^{-j\omega t}\,dt$$

CONTINUOUS-TIME

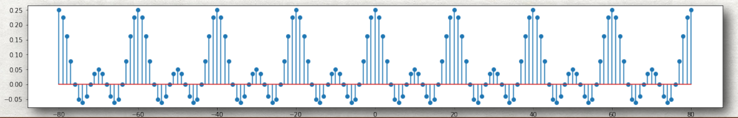
DISCRETE-TIME

FOURIER SERIES

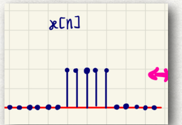


$$x[n] = \sum_{k=\langle N\rangle} a_k e^{jk(\omega_0)n} = \sum_{k=\langle N\rangle} a_k e^{jk(2\pi/N)n}$$

$$a_k = \frac{1}{N}\sum_{k=\langle N\rangle} x[n]e^{-jk\omega_0 n} = \frac{1}{N}\sum_{k=\langle N\rangle} x[n]e^{-jk(2\pi/N)n}$$

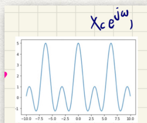


FOURIER TRANSFORM

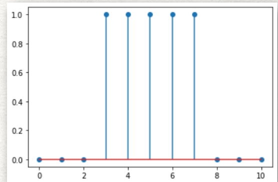


$$x[n] = \frac{1}{2\pi}\int_{2\pi} X(e^{j\omega})e^{j\omega n}\,d\omega$$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$$

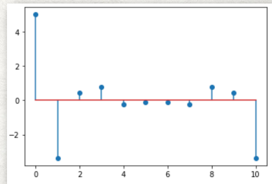


DFT



$$x[n] = \frac{1}{N}\sum_{n=0}^{N-1} X[k]e^{j(2\pi/N)nk}$$

$$X[k] = \sum_{k=0}^{N-1} x[n]e^{-j(2\pi/N)nk}$$



FILTERING

- A process to change the relative amplitudes of the frequency components in a signal or perhaps eliminate some frequency components entirely.

1. **Frequency-shaping filters** - to change the shape of the spectrum.

2. **Frequency-selective filters** - to pass some frequencies essentially undistorted and significantly attenuate or eliminate others.

FREQUENCY-SHAPING FILTERS

Frequency-shaping filters - to change the shape of the spectrum

- (1) **Audio systems** that allow the listener to modify relative amounts of low-frequency energy (bass) and high-frequency energy (treble). The frequency responses of the system can be changed by manipulating tone controls.
  - High-fidelity audio systems, a so-called equalising filter is often included in the preamplifier to compensate for the frequency-response characteristics of the speakers.
  - Three cascaded filtering stages (referred as equalizing or equaliser circuits) shown in log-log plot (magnitude in units of $20\,log_{10}|H(j\omega)|$ referred to as decibels of dB, frequency axis in Hz along a logarithmic scale).

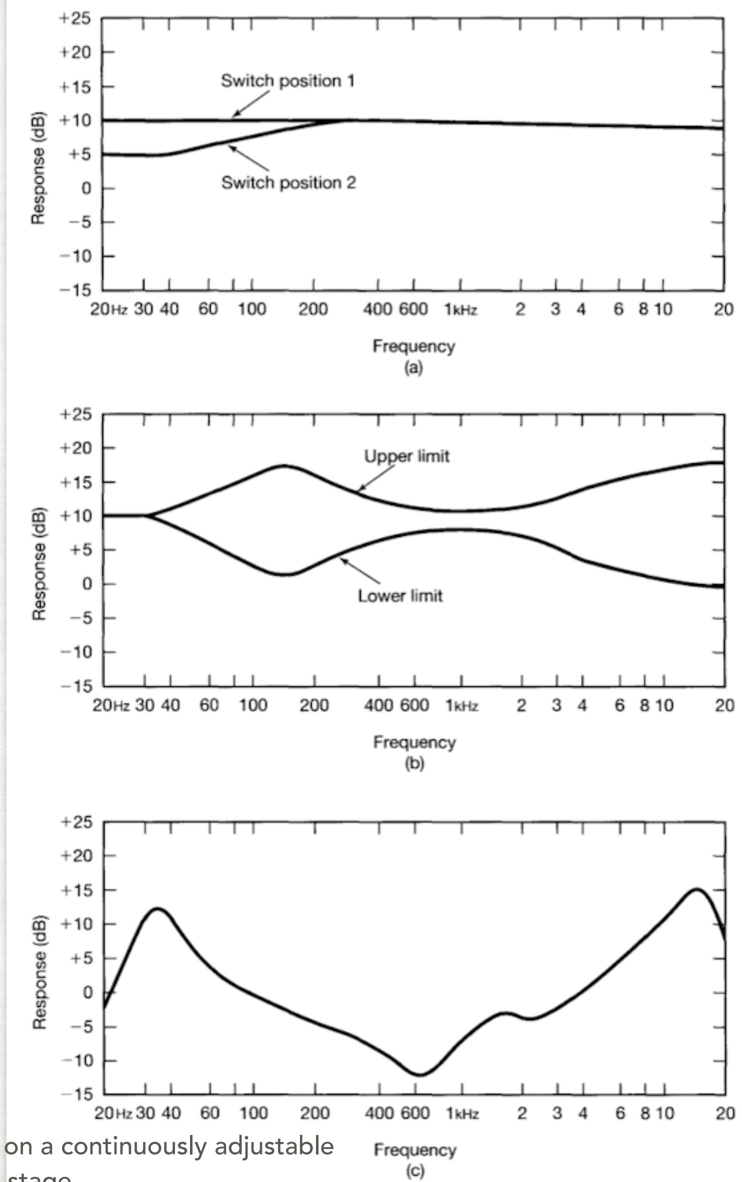
FREQUENCY-SHAPING FILTERS

Frequency-shaping filters - to change the shape of the spectrum

- **Audio systems** - The equalising circuits are designed to compensate for the frequency response of the speakers and the room in which they are located to allow the listener to control the overall frequency response.

- The cascade of the three systems is the product of the three frequency responses.

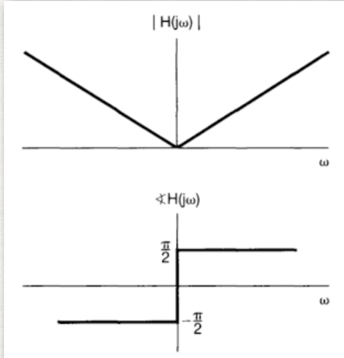
- First two filters make up the control stage of the system. The third one is the equaliser stage which has the fixed frequency response indicated.



(a) low freq. controlled by two position switches (b) upper and lower limits on a continuously adjustable shaping filter (c) frequency response of the equaliser stage.

FREQUENCY-SHAPING FILTERS

$H(j\omega) = j\omega$



Frequency-shaping filters - to change the shape of the spectrum

- (2) **Differencing filter** - filter output is the derivative o the filter input, $y(t) = dx(t)/dt$, frequency response is $H(j\omega) = j\omega$
  - Used for enhancing rapid variations or transitions in a signal, e.g., enhance edges in picture processing.
  - A black-and-white picture - two-dimensional signal $x(t_1, t_2)$, where t_1 and t_2 are the horizontal and vertical coordinates, respectively.

Effect of a differentiating filter on an image (a) original images, (b) the results of processing the image with a differencing filter

FREQUENCY-SHAPING FILTERS

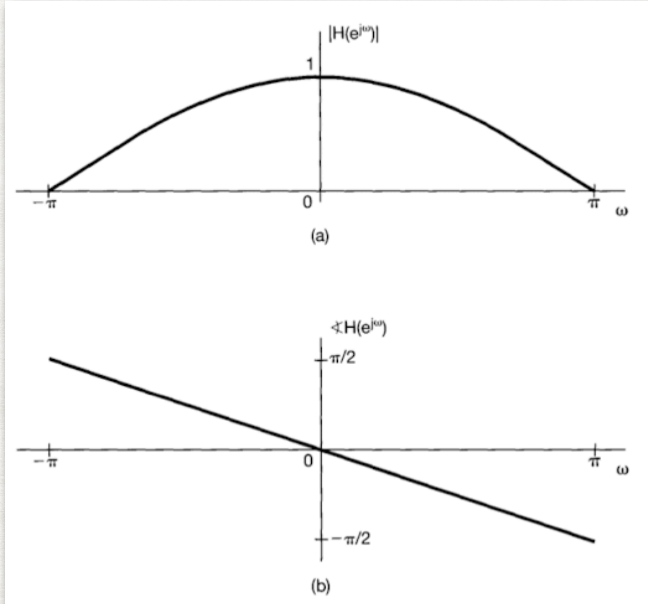
Frequency-shaping filters - to change the shape of the spectrum

- (3) **Two-point averaging filter** (Discrete-time systems)

$$y[n] = \frac{1}{2}(x[n] + x[n-1])$$

In this case, $h[n] = \frac{1}{2}(\delta[n] + \delta[n-1])$

$$H(e^{j\omega}) = \frac{1}{2}(1 - e^{-j\omega}) = e^{-j\omega/2}cos(\omega/2)$$



Magnitude (a) and phase (b) of y[n]

For example, $x[n] = Ke^{j\omega n}$

For low frequency, $x[n] = Ke^{j0n}$ = K

For ω=0, output will be $y[n] = H(e^{j0})Ke^{j0n}$

For high frequency, $x[n] = Ke^{j\pi n}$ = $(-1)^n K$

For ω=π, output will be $y[n] = H(e^{j\pi})Ke^{j\pi n}$

FREQUENCY-SELECTIVE FILTERS

Frequency-selective filters - to pass some frequencies essentially undistorted and significantly attenuate or eliminate others.

- (1) Process by which useful part of a signal is separated from extraneous and undesirable components, generally referred as "noise"
- (2) selective filter in communication systems is for transmission of information from many different sources simultaneously by putting the information from each channel into a separate frequency Band and extracting individual channels or bands at the receiver using frequency-selective filters (to adjust quality using frequency-shaping filters)
- Filtering based on convolution property of Fourier transform - Fourier transform of the output is the product of Fourier transform of the input and the frequency response of the system.
- Ideal frequency-selective filters
  - Pass band : $|H(j\omega)| = 1$ Stop band : $|H(j\omega)| = 0$

FREQUENCY-SELECTIVE FILTERS

Ideal frequency-selective filter passes complex exponentials at one set of frequencies without any distortion and completely rejects signals at all other frequencies.

- Most common types of filters are the following
  - Low-pass filters
  - High-pass filters
  - Band-pass filters
  - Band-stop filters

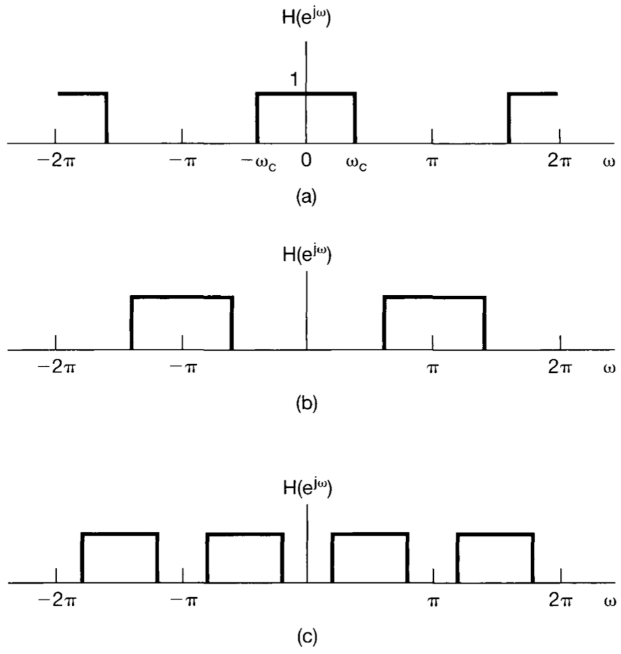


Figure 3.28 Discrete-time ideal frequency-selective filters: (a) lowpass; (b) highpass; (c) bandpass.

Ideal filters are quite useful in describing idealised system, but not realisable in practice and must be approximated. Although, realisable, it might make undesirable for particular applications. A non ideal filter might in fact be preferable.

FREQUENCY-SELECTIVE FILTERS

EXERCISE

- Consider the ideal low-pass filter with frequency response

$$H_{lp}(e^{j\omega}) = \begin{cases} 1, & |\omega| < \omega_c \\ 0, & elsewhere \end{cases}$$

$$H_{hp}(e^{j\omega}) = H_{lp}(e^{j(\omega-\pi)})$$

$$h_{hp}[n] = h_{lp}[n](e^{j\pi n})$$

$$= (-1)^n h_{lp}[n]$$

FILTERING

- Practical Filters
  - Low-pass filters

$$1 - \delta_1 \leq |H(\omega)| \leq 1 + \delta_1, \quad |\omega| \leq \omega_p$$

$$|H(\omega)| \leq \delta_2, \quad |\omega_s| \leq \omega$$

Where ω_p and ω_s are pass band and stop band cutoff frequencies, respectively

The range between ω_p and ω_s are the transition band

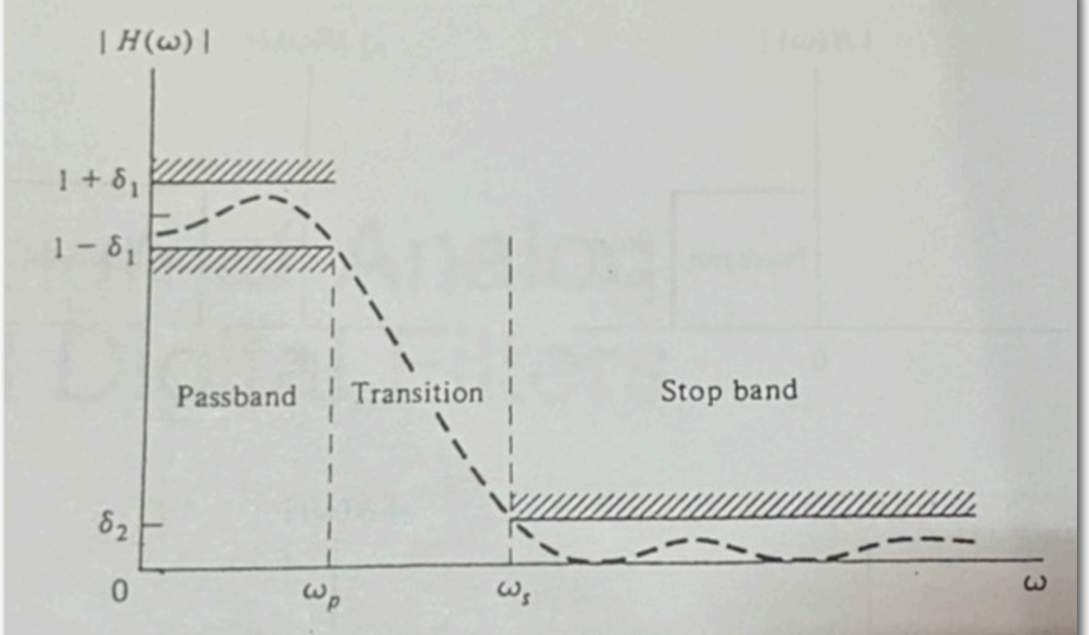
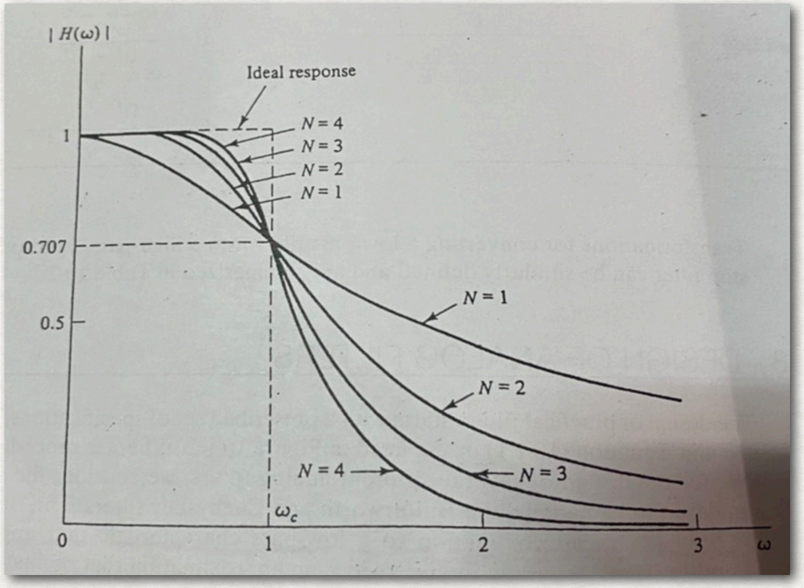


Figure 10.1.3 Specification for practical low-pass filter.

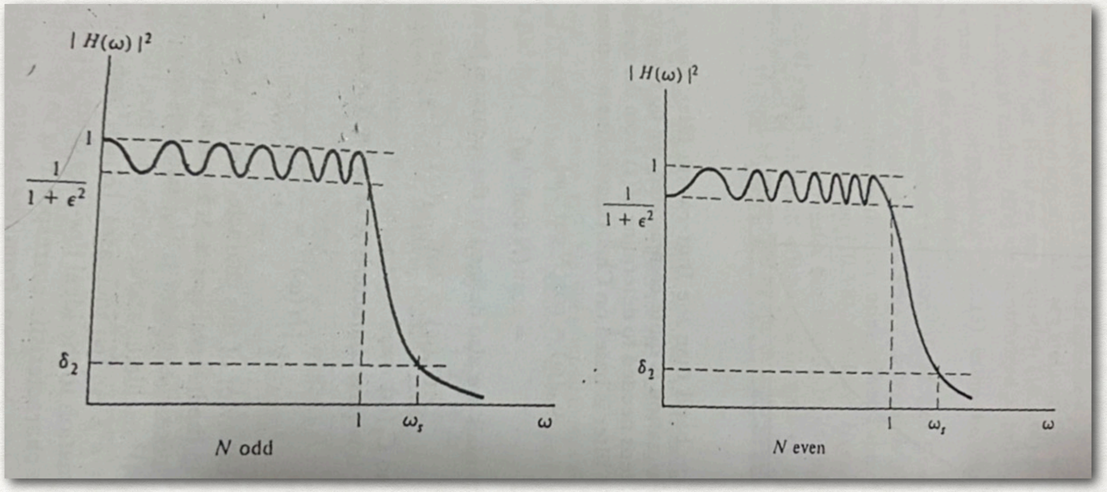
FILTER DESIGN

ANALOG FILTERS

- Analog Filter design, e.g., Butterworth filter - an approximation to a low-pass characteristics that approaches zero smoothly, Chebyshev filter - an approximation that oscillates in the pass-band but monotonically decreases in the transition and stop bands



Magnitude plot of normalised Butterworth

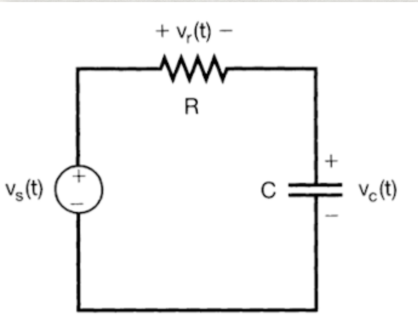


Magnitude characteristic of the Chebyshev approximation

Continuous and discrete signals and systems by S.S. Soliman & M.D. Srinath

FREQUENCY-SELECTIVE FILTERS

SIMPLE RC LOWPASS FILTER



Source voltage $v_s(t)$ - system input

The circuit can be used as a lowpass filter when take capacitor voltage $v_c(t)$ as the system output.

TIME DOMAIN

$$RC\frac{dv_c(t)}{dt} + v_c(t) = v_s(t)$$

$$RC\frac{dh(t)}{dt} + h(t) = \delta(t)$$

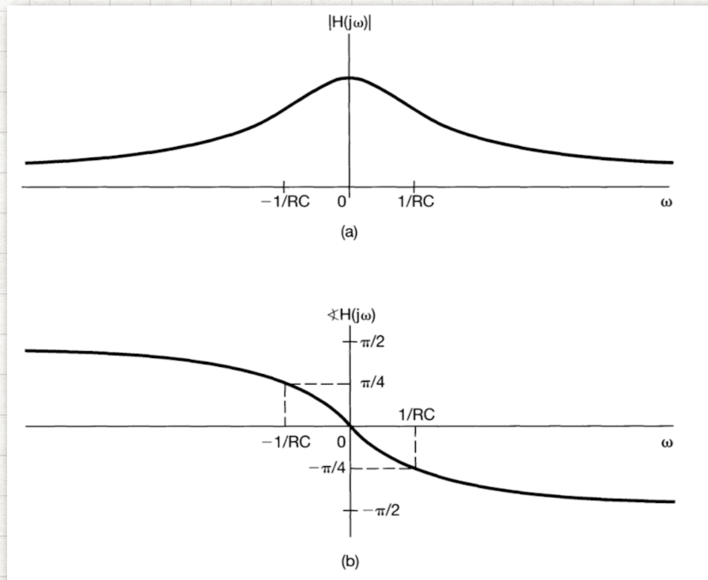
FREQ DOMAIN

$$RCj\omega H(j\omega) + H(j\omega) = 1$$

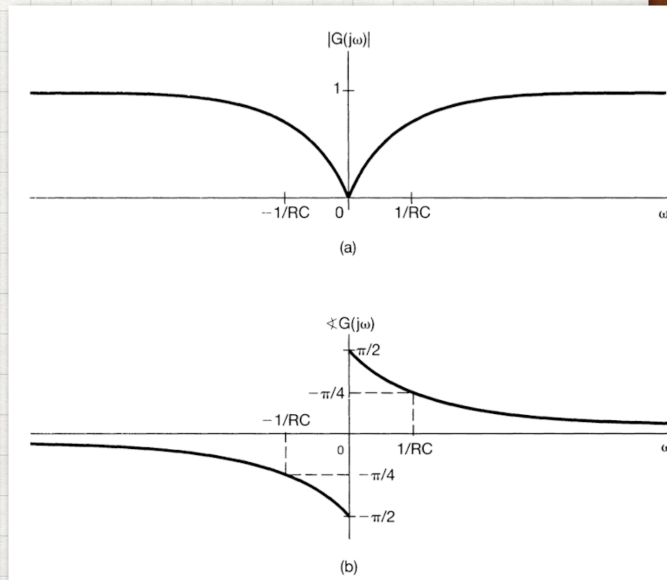
$$H(j\omega) = \frac{1}{1 + RCj\omega}$$

TIME DOMAIN

$$h(t) = \frac{1}{RC}e^{-t/RC}u(t)$$



(1) Magnitude and (2) phase of the RC circuit with $v_c(t)$ output



(1) Magnitude and (2) phase when choosing voltage across resistor $v_r(t)$ to get highness filter

FILTER DESIGN

DIGITAL FILTERS

- Digital Filter design FIR / IIR digital filters

General-purpose digital systems, e.g., difference equations are widely used in practice

1) Recursive and have impulse response of infinite length (IIR)

2) non-recursive and have impulse response of finite length (FIR)

Signals and Systems
by A. V. Oppenheim & A.S. Willsky

FIRST-ORDER RECURSIVE DISCRETE-TIME FILTERS

- First-order filters in discrete-time counterpart in LTI system,

$$y[n] - ay[n-1] = x[n]$$

- if $x[n] = \delta[n]$, then $y[n] = h[n]$, where $H(e^{j\omega})$ is the frequency response of the system, substitute the equation above to get

$$H(e^{j\omega}) - aH(e^{j\omega})e^{-j\omega} = 1$$

$$H(e^{j\omega}) = \frac{1}{1 - ae^{-j\omega}}$$

- The impulse response of the system is

$$h[n] = a^n u[n]$$

- $|a|$ control the speed, with faster response for smaller values of $|a|$

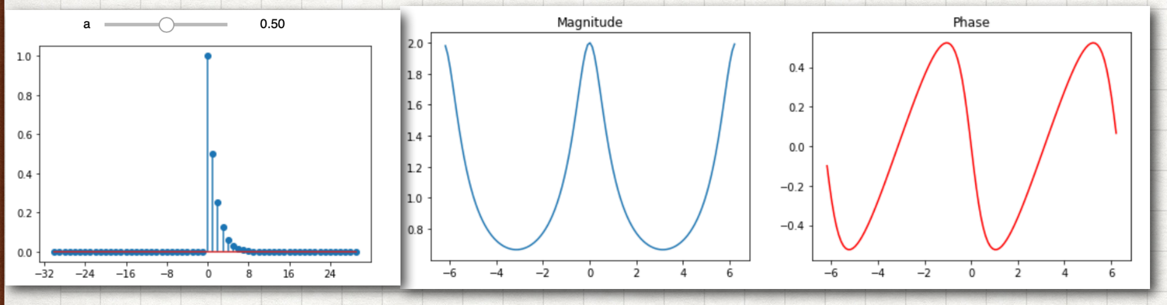
- $|a| > 1$, the system is unstable (we are not considering this).

FIRST-ORDER RECURSIVE DISCRETE-TIME FILTERS

FREQUENCY RESPONSE OF THE FIRST-ORDER RECURSIVE DISCRETE-TIME FILTER

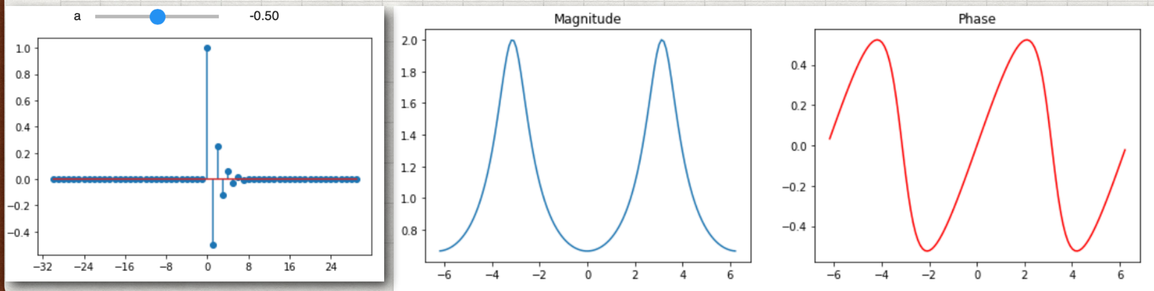
The signal $x[n] = a^n u[n] \quad where\,|a| < 1$

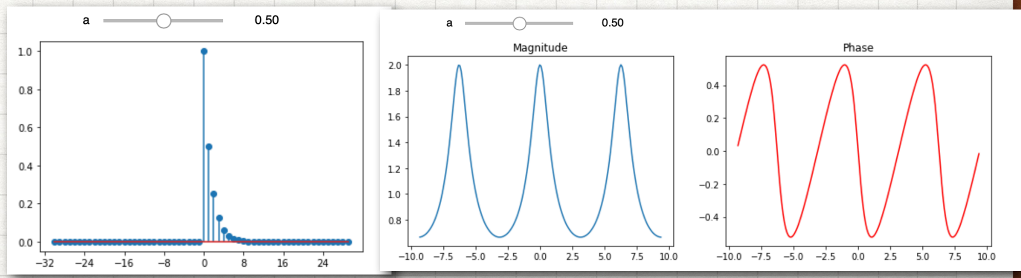
$$X(e^{j\omega}) = \frac{1}{1 - ae^{-j\omega}}$$

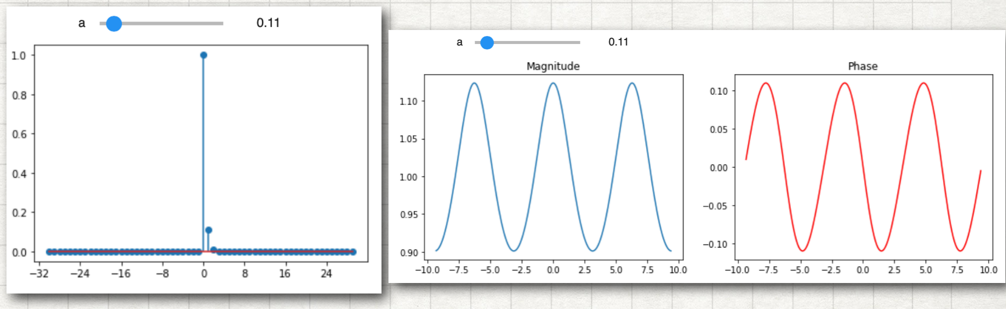


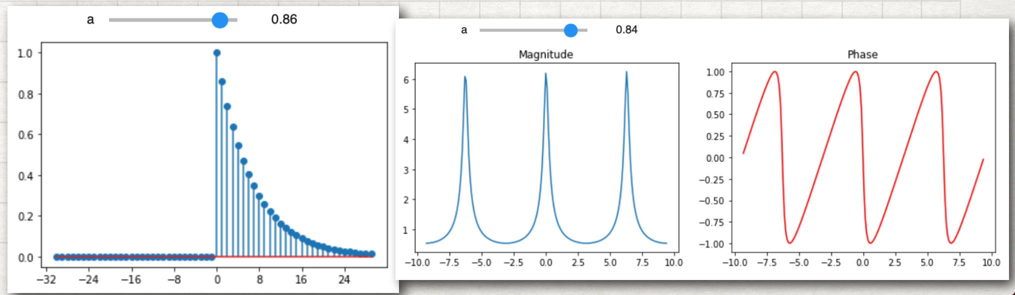
The signal $x[n] = (-1)^n a^n u[n] \quad where\,|a| < 1$

$$X(e^{j\omega}) = \frac{1}{1 + ae^{-j\omega}}$$









NON-RECURSIVE DISCRETE-TIME FILTERS

- The general form of an FIR discrete-time non-recursive equation is

$$y[n] = \sum_{k=-N}^{M} b_k x[n-k]$$

- That is, the output y[n] is a weighted average of the N+M+1 values of x[n] from x[n+N] through x[n-M] with the weights given by the coefficients b_k.

- An example of such a filter is a moving-average filter, where the output y[n] for any n — say, n_0— is an average of values of x[n] in the vicinity of n_0.

    - The basic idea is that by averaging values locating, rapid high-frequency components for the input will be averaged out and lower frequency will be retained.

NON-RECURSIVE DISCRETE-TIME FILTERS

- Three-point moving-average filter, which is of the form

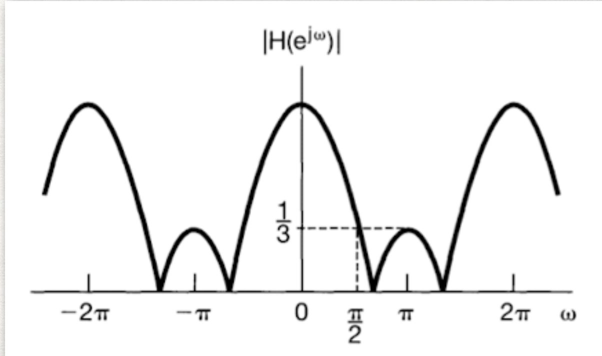
$$y[n] = \frac{1}{3}[x[n-1] + x[n] + x[n+1]]$$

- so we can find impulse response from $x[n] = \delta[n]$

$$h[n] = \frac{1}{3}[\delta[n-1] + \delta[n] + \delta[n+1]]$$

- The corresponding frequency response is

$$H(e^{j\omega}) = \frac{1}{3}[e^{-j\omega} + 1 + e^{j\omega}] = \frac{1}{3}[1 + 2\cos\omega]$$



Magnitude of the frequency response of a three-point moving-average lowpass filter

NON-RECURSIVE DISCRETE-TIME FILTERS

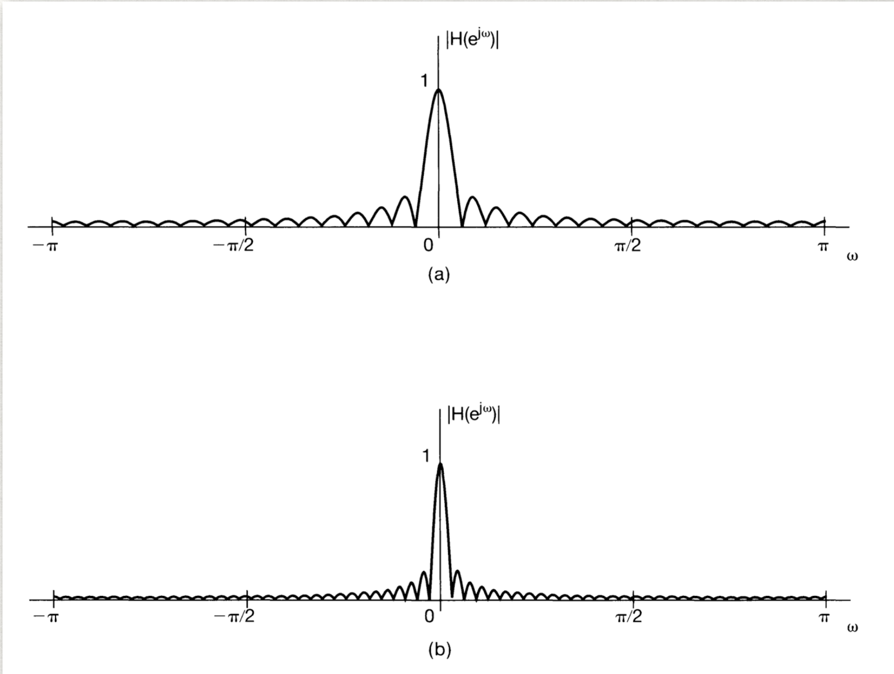
- Consider averaging N+M+1 neighbouring points — use the general form

$$y[n] = \frac{1}{N+M+1} \sum_{k=-N}^{M} x[n-k]$$

- The corresponding impulse response is a rectangular pulse (i.e., $h[n] = 1/(M+N+1)$ for $-N \leq n \leq -M$ and $h[n] = 0$, otherwise. The filter's frequency response is

$$H(e^{j\omega}) = \frac{1}{N+M+1} \sum_{k=-N}^{M} e^{-j\omega k}$$

$$= \frac{1}{N+M+1} e^{j\omega(N-M)/2} \frac{sin[\omega[M+N+1)/2]}{sin(\omega/2)}$$



FIR: Magnitude of the frequency response for the lowpass moving-average filter (a) M = N = 16, (b) M = N = 32

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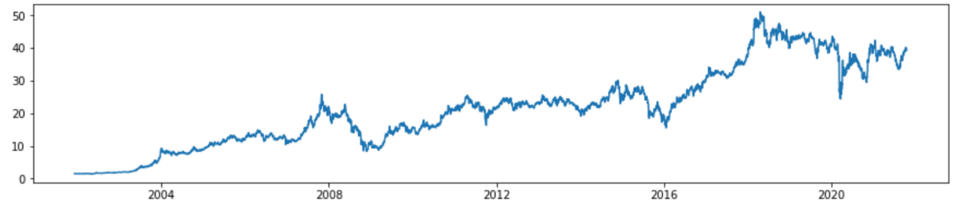
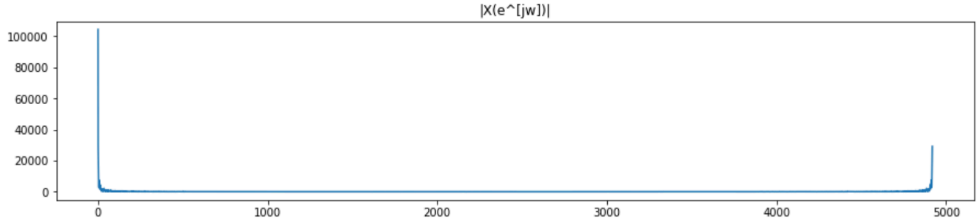
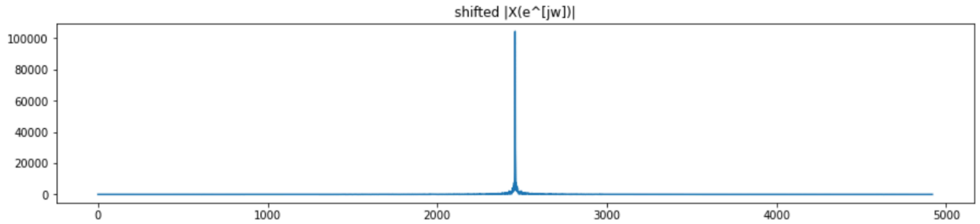
65-POINT MOVING AVERAGE

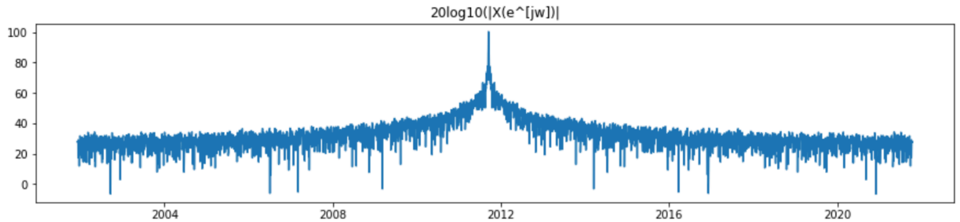
```
Date
2001-12-06      1.553146
2001-12-07      1.542285
2001-12-10      1.542285
2001-12-11      1.477117
2001-12-12      1.520562
                  ...
2021-10-08     39.750000
2021-10-11     40.000000
2021-10-12     40.000000
2021-10-14     39.500000
2021-10-15     39.250000
Name: Adj Close, Length: 4919, dtype: float64
```

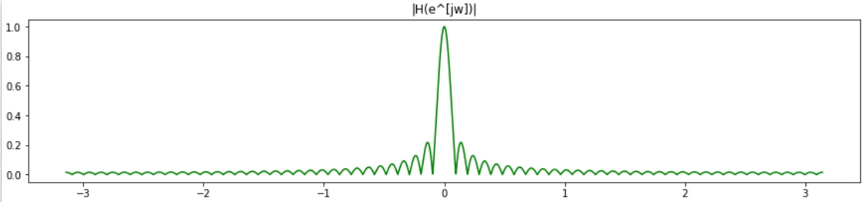


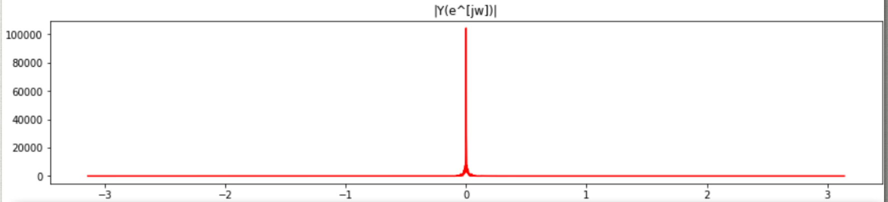


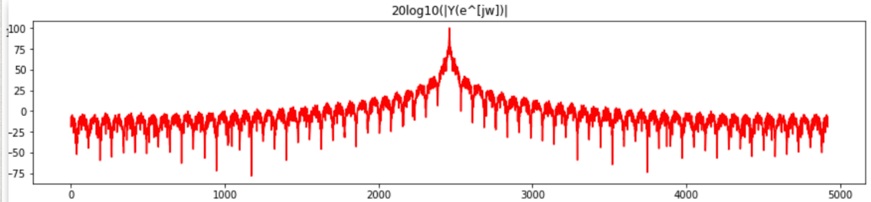


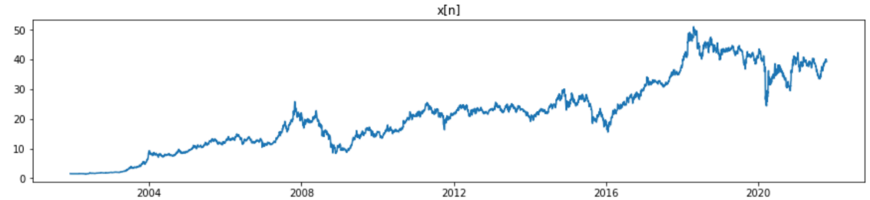

```
(4919,)
```

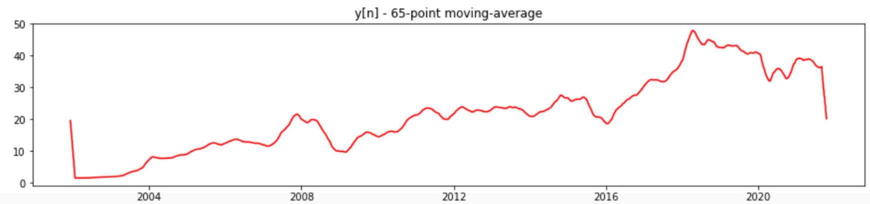




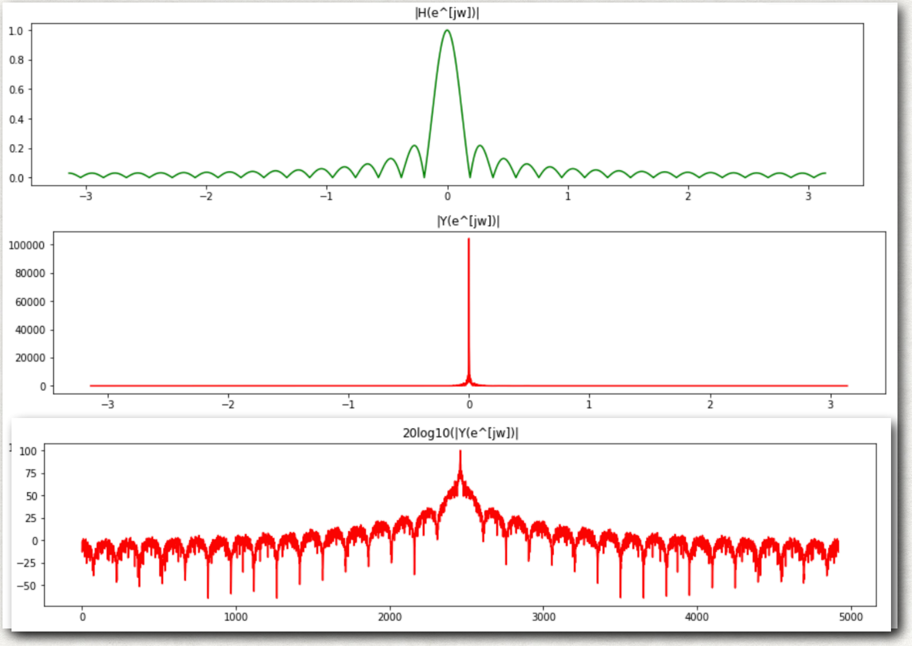


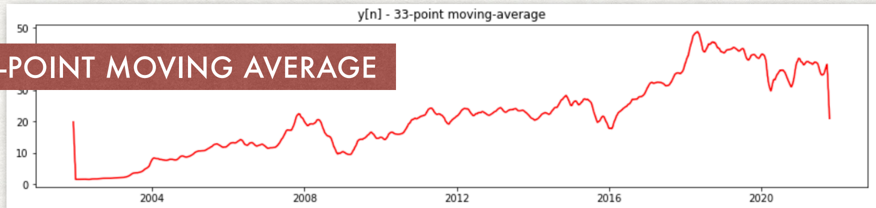




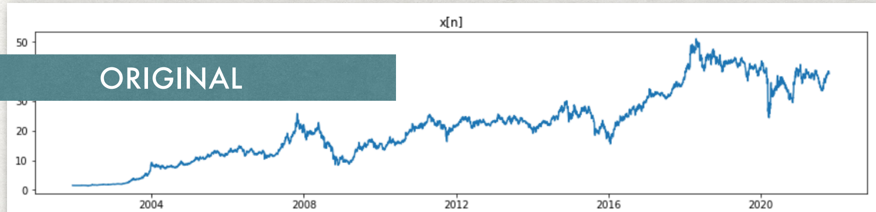


33-POINT MOVING AVERAGE



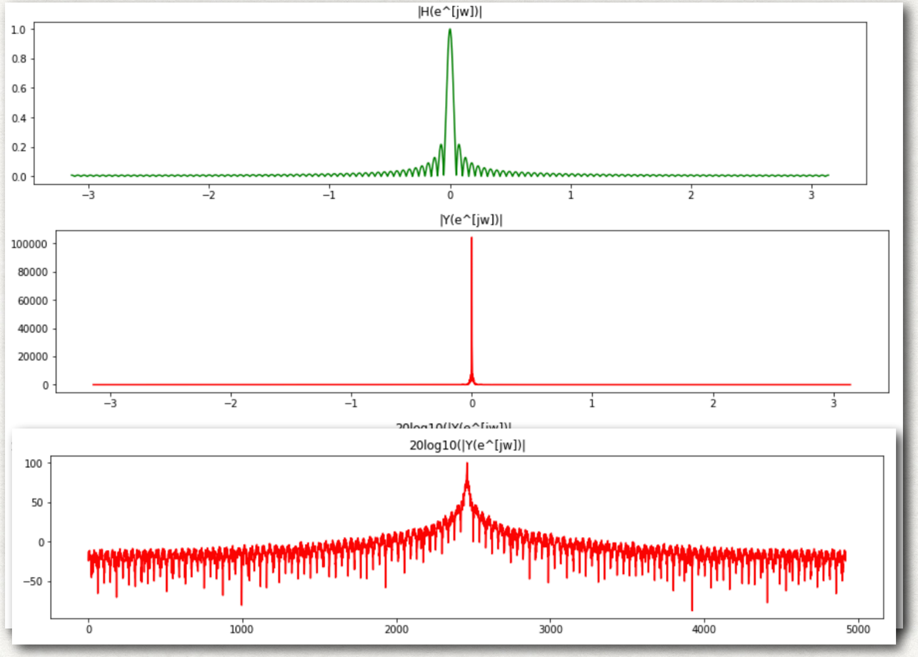


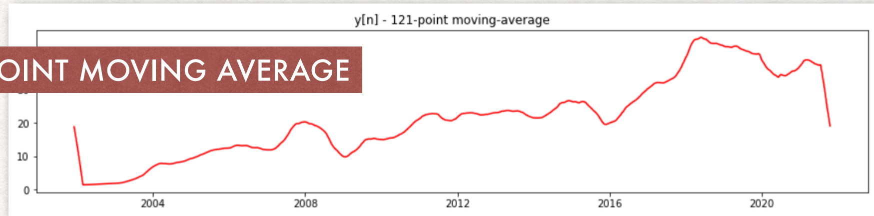
33-POINT MOVING AVERAGE



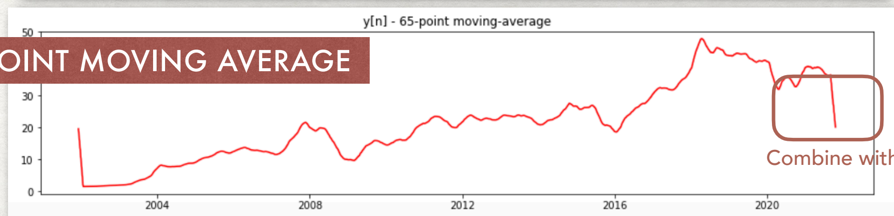
ORIGINAL

121-POINT MOVING AVERAGE





121-POINT MOVING AVERAGE



65-POINT MOVING AVERAGE

Combine with x[n]

COMPARED COMPUTATIONAL TIME

```
(100000,)
0.005743503570556641
0.005539417266845703
0.0053272247314453125
0.005335330963134766
0.005251884460449219
0.004944324493408203
0.005033969879150391
0.004873037338256836
0.005079030990600586
0.0055103302001953125
#####################################
0.002890348434448242
0.0035409927368164062
0.002797842025756836
0.002798318862915039
0.002932310104370117
0.0030007362365722656
0.002913236618041992
0.002914905548095703
0.0029027462005615234
0.0029032230377197266
Datasize: 100000 N-day moving average: 15 days
average time in Freq Domain 0.005263805389404297
average time in Time Domain 0.002959465980529785
```

```
(100000,)
0.005747795104980469
0.006157636642456055
0.0053865909576416016
0.005326747894287109
0.005155801773071289
0.005195140838623047
0.005203723907470703
0.005529880523681641
0.005347013473510742
0.007149696350097656
#####################################
0.005568265914916992
0.005654811859130859
0.005493879318237305
0.005560398101806641
0.005552530288696289
0.005468130111694336
0.0058743953704833984
0.009973287582397461
0.009770870208740234
0.0056455135345458984
Datasize: 100000 N-day moving average: 200 days
average time in Freq Domain 0.005620002746582031
average time in Time Domain 0.006456208229064941
```

```
(100000,)
0.018497467041015625
0.0051708221435546875
0.015355110168457031
0.021502017974853516
0.005101442337036133
0.0053403377532958984
0.005267620086669922
0.015375137329101562
0.005295276641845703
0.005459308624267578
#####################################
0.11014127731323242
0.06126546859741211
0.06128501892089844
0.06788206100463867
0.050124406814575195
0.03345775604248047
0.03833627700805664
0.033537864685058594
0.03812861442565918
0.06925106048583984
Datasize: 100000 N-day moving average: 2000 days
average time in Freq Domain 0.010236454010009766
average time in Time Domain 0.05634098052978516
```

NON-RECURSIVE DISCRETE-TIME FILTERS

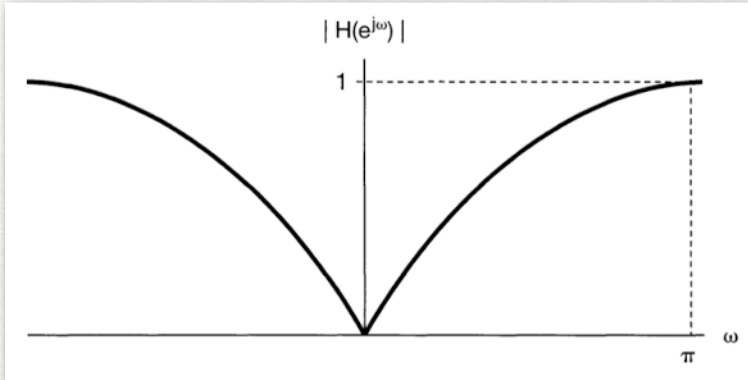
- Non-recursive filters can also be used to perform highness filtering operations. Consider the difference equation,

$$y[n] = \frac{x[n] - x[n-1]}{2}$$

- If input signals are approximately constant, the value of y[n] is close to zero. For inputs that vary greatly from sample to sample, the values of y[n] can have larger amplitude.

- In this case,

$$H(e^{j\omega}) = \frac{1}{2}[1 - e^{-j\omega}] = je^{-j\omega/2}sin(\omega/2)$$



Frequency response of a simple highness filter

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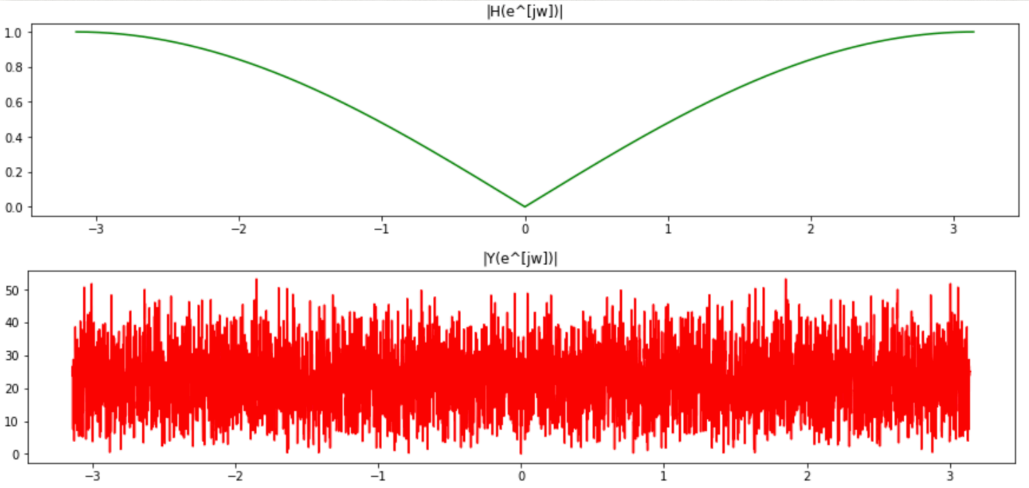
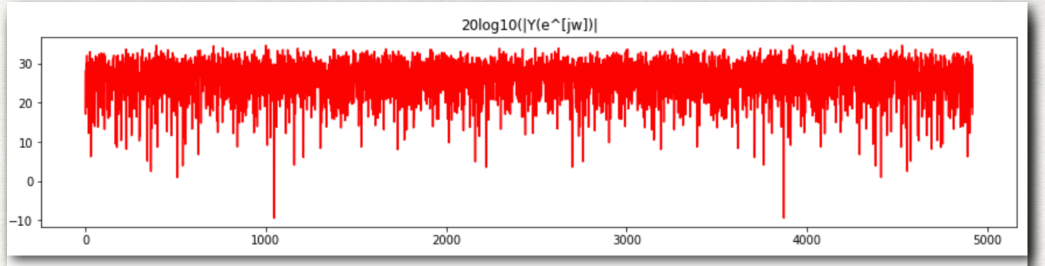
by A. V. Oppenheim & A.S. Willsky

HIGHPASS FILTER





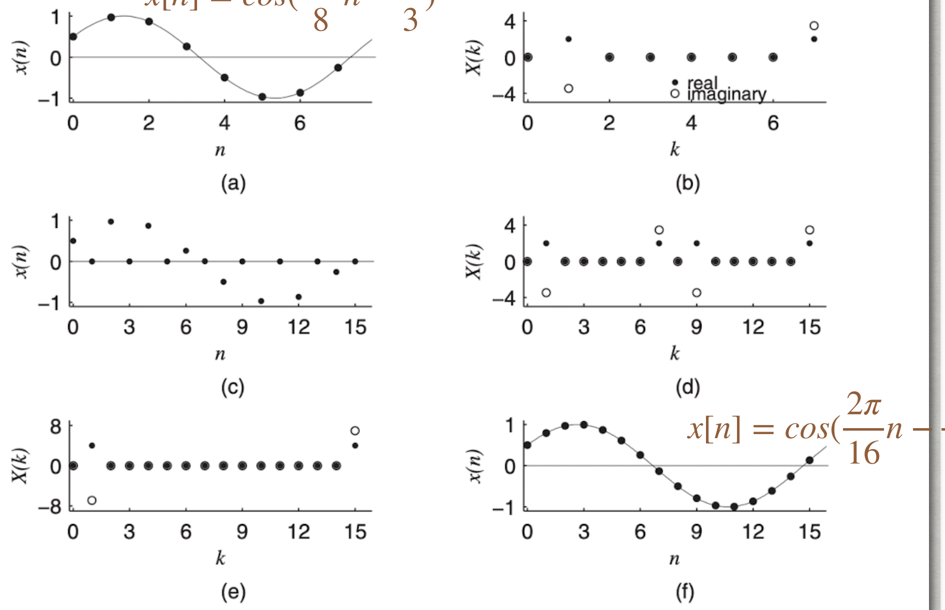
???

OTHER APPLICATIONS

- Interpolation & Decimation using DFT
  - Interpolation - increasing the sampling rate of a signal by factor I (upsampling)
  - Decimation - reducing the sampling rate of a signal by a factor D (downsampling)

$I = 2$

$x[n] = cos(\frac{2\pi}{8}n - \frac{\pi}{3})$



$x[n] = cos(\frac{2\pi}{16}n - \frac{\pi}{3})$

Figure 6.3 (a) A real signal; (b) its spectrum; (c) the signal shown in (a) with zero padding in between the samples; (d) its spectrum, which is the same as that shown in (b), but repeats; (e) the spectrum shown in (d) after lowpass filtering; (f) the corresponding time-domain signal, which is an interpolated version of that shown in (a)

To reduce high frequency

$D = 2$

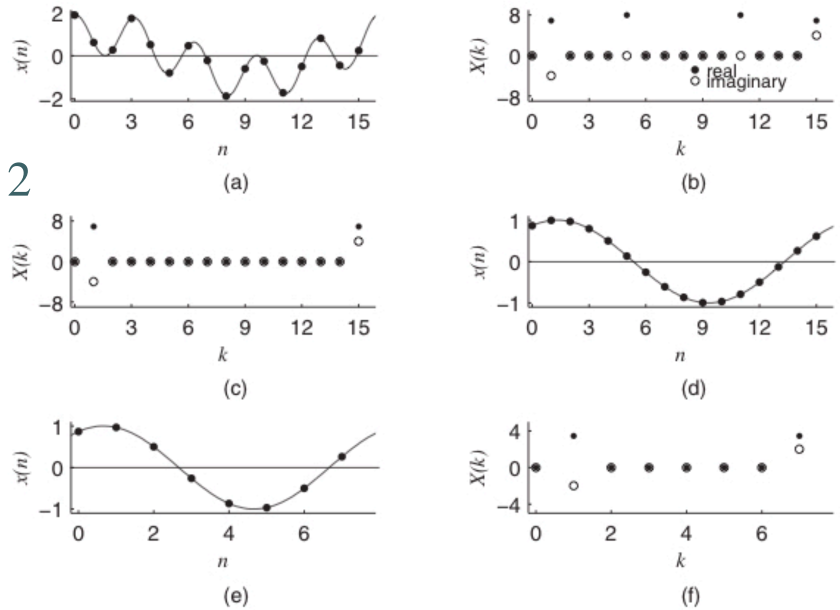


Figure 6.4 (a) A real signal; (b) its spectrum; (c) the spectrum shown in (b) after lowpass filtering; (d) the corresponding time-domain signal; (e) the signal shown in (d) with decimation of alternate samples; (f) its spectrum, which is the same as that shown in (c), but compressed

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OTHER APPLICATIONS

- Interpolation & Decimation $I = 3, D = 2, I/D = 3/2$

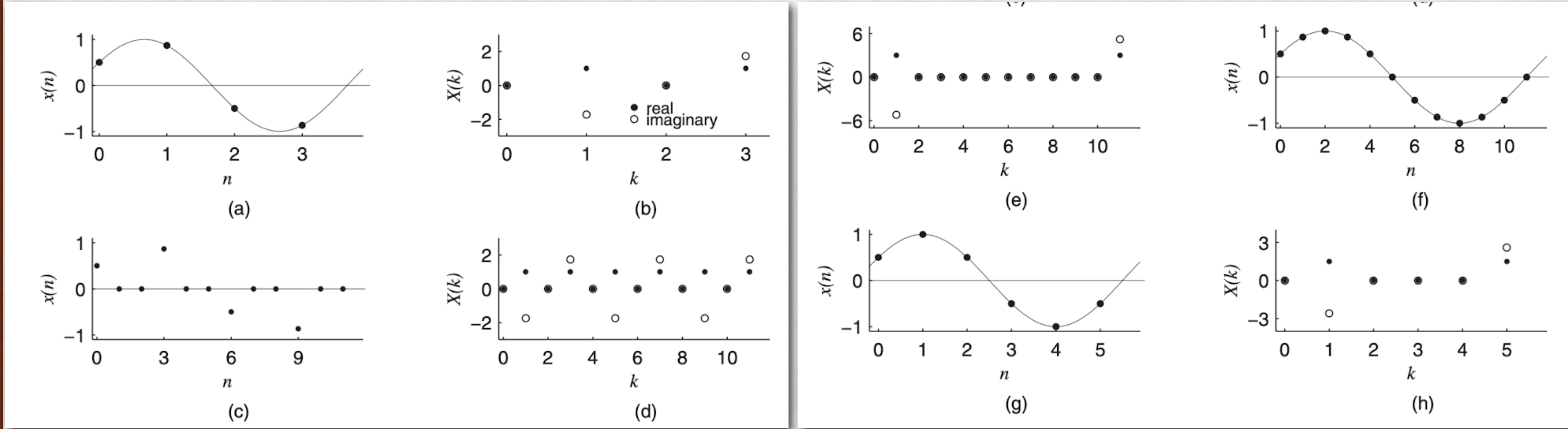

Figure 6.5 (a) A real signal; (b) its spectrum; (c) the signal shown in (a) with zero padding in between the samples; (d) its spectrum, which is the same as that shown in (b), but repeats twice; (e) the spectrum shown in (d) after lowpass filtering; (f) the corresponding time-domain signal, which is an interpolated version of that shown in (a); (g) the signal shown in (f) with decimation of alternate samples; (h) its spectrum, which is the same as that shown in (e), but compressed

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REFERENCES

*****

- Alan V. Oppenheim, Alan S. Willsky and S. Hamid Nawab., **Signals and Systems,** 2nd Edi
  Prentice Hall (1996)
    - Chapter 3 (3.9 - 3.11), Chapter 6, (6.7)
- D. Sundararajan, **A practical approach to Signals and Systems,** Wiley (2008)
    - Section 6.5.2 Interpolation and Decimation